

Problem Set 6

Instructions: Answer the following questions in their entirety in a separate document. Submit your completed problem set as a PDF document via email to `sean.corcoran@vanderbilt.edu`. Use your last name and problem set number as the filename (e.g., *Oliver Problem Set 6.pdf*). Working together is encouraged, but it is expected that all submitted work be that of the individual student.

1. **(12 points—2 each)** Suppose that the historical mean number of days absent among high school students in a state is 8 days per year, with a population standard deviation of 6 days. A researcher suspects that absenteeism has increased. She selects a random sample of records for 400 students and finds that the average number of days absent in the sample is 8.6. Assume that the population standard deviation remains 6 days.
 - (a) Write down H_0 and H_1 in terms of the population mean μ .
 - (b) Describe the approximate sampling distribution of $\bar{x}$ if H_0 is true. What are its mean and standard error?
 - (c) Calculate the appropriate test statistic.
 - (d) Calculate the p -value. Interpret this in words.
 - (e) Test the hypothesis at significance levels $\alpha = 0.05$ and $\alpha = 0.01$. What do you conclude?
 - (f) A school board member interprets your findings as follows: "We have demonstrated that student absenteeism has increased." Is this an appropriate interpretation? Explain.
2. **(6 points)** A researcher tests whether average daily screen time among adolescents exceeds 4 hours per day. She obtains a p -value of 0.032. Consider the following statements. For each, indicate whether it is correct or incorrect, and briefly explain your answer.
 - (a) There is a 3.2% probability that the null hypothesis is true.
 - (b) If the null hypothesis is true, the probability of obtaining a test statistic at least as extreme in the direction of the alternative as the one observed is 3.2%.
 - (c) The researcher should reject the null hypothesis at the 5% significance level.

- (d) The researcher should reject the null hypothesis at the 1% significance level.
 - (e) The researcher has demonstrated that average daily screen time exceeds 4 hours.
 - (f) If the researcher does not reject the null hypothesis, she has established that average screen time is 4 hours.
3. **(8 points–2 each)** Suppose the average reading score among students in a particular state has historically been 500 points, with a known population standard deviation of 80 points. A researcher believes that the mean has declined and plans to test:

$$H_0 : \mu = 500 \quad H_1 : \mu < 500$$

She collects a random sample of 256 students.

- (a) What is the standard error of the sample mean?
 - (b) At a significance level of $\alpha = 0.05$, what is the largest value of $\bar{x}$ for which the researcher would reject H_0 ?
 - (c) Suppose the sample mean is 492. Would you reject H_0 at the 5% level? Explain.
 - (d) If the researcher instead uses $\alpha = 0.01$, would the critical value (as you found in part b) be higher or lower? Explain intuitively before calculating it.
4. **(8 points–2 each)** An education researcher evaluates a new instructional program and reports that students exposed to the program have a higher average test score than a historical benchmark. The researcher obtains a one-sided p -value of 0.048.
- (a) Would the researcher reject H_0 at $\alpha = 0.10$, $\alpha = 0.05$, and $\alpha = 0.01$?
 - (b) Suppose the p -value had instead been 0.052. How would the conclusions change?
 - (c) Is there a meaningful difference in the strength of the statistical evidence between a p -value of 0.048 and a p -value of 0.052?
 - (d) A news article reports that the first study "proved the program works" while the second study "found no evidence that the program works." Critique these characterizations.
5. **(12 points–2 each)** Two researchers are independently investigating whether mean mathematics achievement has increased above a historical average of 500. Both researchers obtain a sample mean of 505. Both assume that the population standard deviation is $\sigma = 100$. However, they use different sample sizes.

	Researcher A	Researcher B
Sample mean	505	505
Population SD	100	100
Sample size	100	2,500

- Calculate the standard error of the sample mean for each researcher.
- Calculate the test statistic and one-sided p -value for each researcher.
- At $\alpha = 0.05$, which researcher rejects $H_0 : \mu = 500$?
- Both researchers estimate the same difference of 5 points. Why do they reach different conclusions?
- Does a statistically significant finding necessarily indicate that the estimated difference is substantively important for education policy? Explain, and speculate on how you might determine whether a difference is “substantively important.”
- Calculate 95% confidence intervals for the population mean using each sample. Compare their widths.